

```

=====
DEBUG TRACE - first 30 transitions
lambda = 5, T = 30, B = 0, seed = 1
=====
Step 1 | state= 0 | event=arrival (accepted) | arrivals= 1 | new state= 1
Step 2 | state= 1 | event=departure           | arrivals= 1 | new state= 0
Step 3 | state= 0 | event=arrival (accepted) | arrivals= 2 | new state= 1
Step 4 | state= 1 | event=departure           | arrivals= 2 | new state= 0
Step 5 | state= 0 | event=arrival (accepted) | arrivals= 3 | new state= 1
Step 6 | state= 1 | event=arrival (accepted) | arrivals= 4 | new state= 2
Step 7 | state= 2 | event=arrival (accepted) | arrivals= 5 | new state= 3
Step 8 | state= 3 | event=arrival (accepted) | arrivals= 6 | new state= 4
Step 9 | state= 4 | event=arrival (accepted) | arrivals= 7 | new state= 5
Step 10 | state= 5 | event=arrival (accepted) | arrivals= 8 | new state= 6
Step 11 | state= 6 | event=arrival (accepted) | arrivals= 9 | new state= 7
Step 12 | state= 7 | event=departure           | arrivals= 9 | new state= 6
Step 13 | state= 6 | event=arrival (accepted) | arrivals=10 | new state= 7
Step 14 | state= 7 | event=arrival (accepted) | arrivals=11 | new state= 8
Step 15 | state= 8 | event=arrival (accepted) | arrivals=12 | new state= 9
Step 16 | state= 9 | event=arrival (accepted) | arrivals=13 | new state=10
Step 17 | state=10 | event=arrival (rejected) | arrivals=14 | new state=10
Step 18 | state=10 | event=arrival (rejected) | arrivals=15 | new state=10
Step 19 | state=10 | event=arrival (rejected) | arrivals=16 | new state=10
Step 20 | state=10 | event=departure           | arrivals=16 | new state= 9
Step 21 | state= 9 | event=arrival (accepted) | arrivals=17 | new state=10
Step 22 | state=10 | event=arrival (rejected) | arrivals=18 | new state=10
Step 23 | state=10 | event=departure           | arrivals=18 | new state= 9
Step 24 | state= 9 | event=departure           | arrivals=18 | new state= 8
Step 25 | state= 8 | event=departure           | arrivals=18 | new state= 7
Step 26 | state= 7 | event=arrival (accepted) | arrivals=19 | new state= 8
Step 27 | state= 8 | event=arrival (accepted) | arrivals=20 | new state= 9
Step 28 | state= 9 | event=arrival (accepted) | arrivals=21 | new state=10
Step 29 | state=10 | event=departure           | arrivals=21 | new state= 9
Step 30 | state= 9 | event=departure           | arrivals=21 | new state= 8

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End of debug trace.
To disable debugging for the normal executions, set:
DEBUG_TRACE = false

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=====
Running simulation for lambda = 1, T = 100000, B = 5000
=====
lambda = 1, mu = 5, K = 10, T = 100000, B = 5000
Runtime = 6.6001 sec

--- Using all transitions ---
Mean number in system L      = 0.2511296521
Rejection probability P_rej   = 0.0000000000
Mean time in system W        = 0.2505772043
State probabilities P        = [ 0.79945865 0.16008330 0.03232467 0.00655883
0.00122766 0.00027
151 0.00007539 0.00000000 0.00000000 0.00000000 0.00000000 ]

--- Ignoring first B transitions ---
Mean number in system L_B    = 0.2509471723
Rejection probability P_rej_B = 0.0000000000
Mean time in system W_B      = 0.2506592621
State probabilities P_B      = [ 0.79959641 0.15997915 0.03231434 0.00654240
0.00120567 0.00028
276 0.00007928 0.00000000 0.00000000 0.00000000 0.00000000 ]

Relative difference in L      = 0.072664 %
Relative difference in P_rej  = 0.000000 %
Relative difference in W      = 0.032748 %

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=====
Running simulation for lambda = 5, T = 100000, B = 5000
=====
lambda = 5, mu = 5, K = 10, T = 100000, B = 5000
Runtime = 6.9150 sec

--- Using all transitions ---
Mean number in system L      = 5.0669430209
Rejection probability P_rej   = 0.0955210238
Mean time in system W        = 1.1218601440
State probabilities P         = [ 0.08966699 0.08890558 0.08922529 0.08838208
0.08898398 0.09082
476 0.09041571 0.09071322 0.09425913 0.09341150 0.09521176 ]

--- Ignoring first B transitions ---
Mean number in system L_B    = 5.1071053828
Rejection probability P_rej_B = 0.0968930911
Mean time in system W_B      = 1.1319838096
State probabilities P_B       = [ 0.08845269 0.08741447 0.08785467 0.08707792
0.08838430 0.09068
520 0.09099106 0.09164490 0.09596627 0.09469143 0.09683710 ]

Relative difference in L      = 0.792635 %
Relative difference in P_rej  = 1.436403 %
Relative difference in W      = 0.902400 %

=====
Running simulation for lambda = 10, T = 100000, B = 5000
=====
lambda = 10, mu = 5, K = 10, T = 100000, B = 5000
Runtime = 5.8064 sec

--- Using all transitions ---
Mean number in system L      = 9.0373462585
Rejection probability P_rej   = 0.5063846947
Mean time in system W        = 1.8266043984
State probabilities P         = [ 0.00081372 0.00144580 0.00216852 0.00373982
0.00677244 0.01378
517 0.02777216 0.06007758 0.12141213 0.25427131 0.50774136 ]

--- Ignoring first B transitions ---
Mean number in system L_B    = 9.0350260323
Rejection probability P_rej_B = 0.5063205560
Mean time in system W_B      = 1.8278095613
State probabilities P_B       = [ 0.00082343 0.00149152 0.00226441 0.00376726
0.00672410 0.01396
358 0.02808737 0.05998343 0.12131367 0.25374031 0.50784093 ]

Relative difference in L      = 0.025674 %
Relative difference in P_rej  = 0.012666 %
Relative difference in W      = 0.065978 %

=====
Running simulation for lambda = 1, T = 1000000, B = 20000
=====
lambda = 1, mu = 5, K = 10, T = 1000000, B = 20000
Runtime = 65.9007 sec

--- Using all transitions ---
Mean number in system L      = 0.2505027289
Rejection probability P_rej   = 0.0000000000
Mean time in system W        = 0.2503982626
State probabilities P         = [ 0.79980439 0.15989668 0.03226471 0.00646411

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0.00125921 0.00023
789 0.00005671 0.00001337 0.00000214 0.00000078 0.00000000 ]

--- Ignoring first B transitions ---
Mean number in system L_B      = 0.2503824648
Rejection probability P_rej_B   = 0.0000000000
Mean time in system W_B        = 0.2504195813
State probabilities P_B         = [ 0.79990402 0.15981203 0.03225954 0.00645092
0.00126277 0.00023
721 0.00005691 0.00001364 0.00000218 0.00000079 0.00000000 ]

Relative difference in L        = 0.048009 %
Relative difference in P_rej    = 0.000000 %
Relative difference in W        = 0.008514 %

=====
Running simulation for lambda = 5, T = 1000000, B = 20000
=====
lambda = 5, mu = 5, K = 10, T = 1000000, B = 20000
Runtime = 67.4362 sec

--- Using all transitions ---
Mean number in system L        = 5.0026245648
Rejection probability P_rej     = 0.0898944867
Mean time in system W          = 1.1012411221
State probabilities P           = [ 0.08999344 0.09049205 0.09038443 0.09076960
0.09155982 0.09187
900 0.09182116 0.09179528 0.09157453 0.08997642 0.08975426 ]

--- Ignoring first B transitions ---
Mean number in system L_B      = 5.0080060496
Rejection probability P_rej_B   = 0.0901637587
Mean time in system W_B        = 1.1026525045
State probabilities P_B         = [ 0.08983444 0.09029272 0.09023266 0.09062306
0.09149535 0.09182
429 0.09177938 0.09192607 0.09181162 0.09017976 0.09000064 ]

Relative difference in L        = 0.107573 %
Relative difference in P_rej    = 0.299542 %
Relative difference in W        = 0.128163 %

=====
Running simulation for lambda = 10, T = 1000000, B = 20000
=====
lambda = 10, mu = 5, K = 10, T = 1000000, B = 20000
Runtime = 59.6151 sec

--- Using all transitions ---
Mean number in system L        = 9.0193533021
Rejection probability P_rej     = 0.5019511910
Mean time in system W          = 1.8090288162
State probabilities P           = [ 0.00047210 0.00093474 0.00183579 0.00375632
0.00744585 0.01515
244 0.03045462 0.06143927 0.12444488 0.25106915 0.50299484 ]

--- Ignoring first B transitions ---
Mean number in system L_B      = 9.0198466095
Rejection probability P_rej_B   = 0.5021108010
Mean time in system W_B        = 1.8096164018
State probabilities P_B         = [ 0.00046298 0.00089190 0.00181307 0.00376460
0.00744556 0.01519
379 0.03049994 0.06138163 0.12443250 0.25098774 0.50312628 ]

Relative difference in L        = 0.005469 %

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Relative difference in P_rej = 0.031798 %
Relative difference in W = 0.032481 %

=====
Running simulation for lambda = 1, T = 10000000, B = 120000
=====

lambda = 1, mu = 5, K = 10, T = 10000000, B = 120000
Runtime = 635.6335 sec

--- Using all transitions ---

Mean number in system L = 0.2504273515
Rejection probability P_rej = 0.0000000000
Mean time in system W = 0.2502386518
State probabilities P = [0.79975162 0.16010075 0.03211215 0.00643726
0.00127917 0.00025
538 0.00005141 0.00001010 0.00000168 0.00000048 0.00000000]

--- Ignoring first B transitions ---

Mean number in system L_B = 0.2504118097
Rejection probability P_rej_B = 0.0000000000
Mean time in system W_B = 0.2502303841
State probabilities P_B = [0.79976174 0.16009561 0.03210828 0.00643548
0.00127993 0.00025
533 0.00005124 0.00001020 0.00000170 0.00000048 0.00000000]

Relative difference in L = 0.006206 %
Relative difference in P_rej = 0.000000 %
Relative difference in W = 0.003304 %

=====
Running simulation for lambda = 5, T = 10000000, B = 120000
=====

lambda = 5, mu = 5, K = 10, T = 10000000, B = 120000
Runtime = 639.4556 sec

--- Using all transitions ---

Mean number in system L = 5.0076954577
Rejection probability P_rej = 0.0911728040
Mean time in system W = 1.1017529664
State probabilities P = [0.09040039 0.09059619 0.09068113 0.09101954
0.09087455 0.09094
612 0.09094979 0.09098016 0.09122787 0.09117577 0.09114848]

--- Ignoring first B transitions ---

Mean number in system L_B = 5.0075222213
Rejection probability P_rej_B = 0.0911554245
Mean time in system W_B = 1.1016549734
State probabilities P_B = [0.09039147 0.09060094 0.09068645 0.09103910
0.09088581 0.09093
975 0.09095386 0.09098918 0.09121022 0.09117228 0.09113094]

Relative difference in L = 0.003459 %
Relative difference in P_rej = 0.019062 %
Relative difference in W = 0.008894 %

=====
Running simulation for lambda = 10, T = 10000000, B = 120000
=====

lambda = 10, mu = 5, K = 10, T = 10000000, B = 120000
Runtime = 626.7589 sec

--- Using all transitions ---

Mean number in system L = 9.0065989686
Rejection probability P_rej = 0.5004547366

Mean time in system W = 1.8020232187
State probabilities P = [0.00046642 0.00096135 0.00192156 0.00390119
0.00784720 0.01562
832 0.03114148 0.06239525 0.12498207 0.25046324 0.50029192]

--- Ignoring first B transitions ---

Mean number in system L_B = 9.0062655890
Rejection probability P_rej_B = 0.5003975920
Mean time in system W_B = 1.8017936156
State probabilities P_B = [0.00046348 0.00095733 0.00192049 0.00390357
0.00785770 0.01564
922 0.03117419 0.06242214 0.12502019 0.25039893 0.50023277]

Relative difference in L = 0.003702 %
Relative difference in P_rej = 0.011419 %
Relative difference in W = 0.012741 %